Exercises Week 6

Advanced Machine Learning (02460)
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1 Theoretical exercises

Exercise 6.1 (Quadratic metric) Consider a two-dimensional abstract manifold with the metric

$$\mathbf{G}_{\mathbf{x}} = \left(1 + \|\mathbf{x}\|^2\right)\mathbf{I}, \qquad \mathbf{x} \in \mathbb{R}^2. \tag{1}$$

Consider the points

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \mathbf{x}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \qquad \mathbf{x}_3 = \begin{pmatrix} 0 \\ 3 \end{pmatrix}. \tag{2}$$

1. *Compute the local norms of the tangent vector* $\mathbf{v}_1 = \begin{pmatrix} 1 & 0 \end{pmatrix}^\intercal$ *assuming the point of tangency is* $\mathbf{x}_1$, $\mathbf{x}_2$ *and* $\mathbf{x}_3$, *respectively.*
2. *Compute the local angles between* $\mathbf{v}_1$ *and* $\mathbf{v}_2 = \begin{pmatrix} 0 & 1 \end{pmatrix}^\intercal$ *in the same three points of tangency.*

Exercise 6.2 (Euclidean metric) Consider the Euclidean metric of $\mathbb{R}^d$, i.e. $\mathbf{G} = \mathbf{I}$.

1. *Derive the coefficients of the geodesics* ODE *of this metric (Eq. 7.20 in the LMLG book).*
2. *Derive the geodesic* ODE.
3. *What is the geodesic that connects points* $\mathbf{x}_1$ *and* $\mathbf{x}_2$*?*

Exercise 6.3 (Quadratic metric) Consider a two-dimensional abstract manifold with the metric

$$\mathbf{G}_{\mathbf{x}} = \left(1 + \|\mathbf{x}\|^2\right)\mathbf{I}, \qquad \mathbf{x} \in \mathbb{R}^2. \tag{18}$$

1. *Derive the coefficients of the geodesics* ODE *of this metric (Eq. 7.20 in the LMLG book). Hint: note that some terms of the metric are zero, which renders some coefficients to be zero as well.*

2. *Derive the geodesic* ODE.

3. *Consider a geodesic c starting at $c_0 = \mathbf{0}$ and initial velocity $\dot{c}_0 = \mathbf{v}$. What is the acceleration $\ddot{c}_0$?*

2 Programming exercises

Exercise 6.4 (Curve parametrizations) Consider a two-dimensional abstract manifold with the metric

$$\mathbf{G}_{\mathbf{x}} = \left(1 + \|\mathbf{x}\|^2\right)\mathbf{I}, \qquad \mathbf{x} \in \mathbb{R}^2. \tag{32}$$

1. *Implement direct energy minimization for computing geodesics using piecewise straight lines to parameterize the solution curve.*

2. *Extend the previous implementation to also support third-order polynomials to parametrize the solution curve.*

Exercise 6.5 (Density metrics) Consider the dataset available at

https://www2.compute.dtu.dk/~sohau/weekendwithbernie/toybanana.npy

This consist of $N = 992$ observations in $\mathbb{R}^2$. Consider the metric over $\mathbb{R}^2$ defined as

$$\mathbf{G}_{\mathbf{x}} = \frac{1}{p(\mathbf{x}) + \epsilon} \tag{33}$$

$$p(\mathbf{x}) = \frac{1}{N}\sum_{n=1}^{N} \mathcal{N}(\mathbf{x}|\mathbf{x}_n, \sigma^2\mathbf{I}), \tag{34}$$

where $\sigma = 0.1$, and $\epsilon = 10^{-4}$ avoids dividing by zero.

1. *Implement direct energy minimization for computing geodesics using piecewise straight lines to parameterize the solution curve.*

2. *Extend the previous implementation to also support third-order polynomials to parametrize the solution curve.*